

Computational Dynamics Q assignment 1

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1. Introduction

This report covers assignment Q1 in Computational Dynamics E25 at Aarhus University. There is an accompanying jupyter notebook in python with the code.

2. Single transformation

To get the rotation matrix from the quaternions requires usage of eq. B.14 [1]

$$\mathbb{F}\mathfrak{R}_{\mathbb{G}}(\mathbb{F}\underline{q}_{\mathbb{G}}) = (q_0\mathbf{I}_3 + \tilde{q})^2 + qq^T \quad (1)$$

The homogenous transform is described by eq. 1.3 [1]

$$\mathbb{F}\mathbb{T}_{\mathbb{G}} \triangleq \begin{bmatrix} \mathbb{F}\mathfrak{R}_{\mathbb{G}} & \mathbb{F}\mathfrak{l}(\mathbb{F}, \mathbb{G}) \\ 0 & 1 \end{bmatrix} \quad (2)$$

With $\mathbb{F}\mathfrak{l}(\mathbb{F}, \mathbb{G})$ being the length from $\mathbb{F}$ to $\mathbb{G}$ as seen from $\mathbb{F}$.

The position of a point in one coordinate frame can be expressed in another coordinate frame by using eq. 1.4 [1]:

$$\begin{bmatrix} \mathbb{F}\mathfrak{p}(\mathbb{F}, \mathbb{O}) \\ 1 \end{bmatrix} = \mathbb{F}\mathbb{T}_{\mathbb{G}} \begin{bmatrix} \mathbb{G}\mathfrak{p}(\mathbb{G}, \mathbb{O}) \\ 1 \end{bmatrix} \quad (3)$$

In the current case eq. 1.2 [1] will be used instead to represent the vector from the point w to z in the reference frame $\mathbb{B}_1$ as the points x and y are coincident:

$$\mathbb{B}_1\mathfrak{l}(w, z) = \mathbb{B}_1\mathfrak{l}(w, x) + \mathbb{B}_1\mathfrak{R}_{\mathbb{B}_2}\mathbb{B}_2\mathfrak{l}(y, z) \quad (4)$$

A skew-symmetric matrix $\tilde{l}$ from a vector l is defined from eq. 1.9 [1]:

$$\tilde{l} \triangleq \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}, \quad l \triangleq \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad (5)$$

The rigid body transform from eq. 1.29 and 1.31 is [1]:

$$\phi(x, y) = \begin{bmatrix} \mathbf{I}_3 & {}^{\mathbb{F}}\tilde{l}(x, y) \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} {}^{\mathbb{F}}\mathfrak{R}_{\mathbb{G}} & \mathbf{0}_3 \\ \mathbf{0}_3 & {}^{\mathbb{F}}\mathfrak{R}_{\mathbb{G}} \end{bmatrix} \quad (6)$$

To then get the velocities in ${}^{\mathbb{I}}\mathcal{V}(\mathbb{I}, z)$ from ${}^{\mathbb{B}_1}\mathcal{V}(\mathbb{I}, w)$ eq. (5) and (6) are used with eq. 1.33 [1]:

$${}^{\mathbb{F}}\mathcal{V}(y) = \phi(x, y)^T {}^{\mathbb{G}}\mathcal{V}(x) \quad (7)$$

Which is used with eq. (4) to transform from point w to z :

$${}^{\mathbb{B}_1}\mathcal{V}(\mathbb{I}, z) = \phi(w, z)^T {}^{\mathbb{B}_1}\mathcal{V}(\mathbb{I}, w) = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 \\ -{}^{\mathbb{B}_1}\tilde{l}(w, z) & \mathbf{I}_3 \end{bmatrix} {}^{\mathbb{B}_1}\mathcal{V}(\mathbb{I}, w) \quad (8)$$

As a last step the velocity is rotated from $\mathbb{B}_1$ to match the inertial fixed reference frame $\mathbb{I}$ which due to only a change in orientation reduces eq. (7) to:

$${}^{\mathbb{I}}\mathcal{V}(\mathbb{I}, z) = \begin{bmatrix} {}^{\mathbb{I}}\mathfrak{R}_{\mathbb{B}_1} & \mathbf{0}_3 \\ \mathbf{0}_3 & {}^{\mathbb{I}}\mathfrak{R}_{\mathbb{B}_1} \end{bmatrix} {}^{\mathbb{B}_1}\mathcal{V}(\mathbb{I}, z) \approx \boxed{\begin{bmatrix} 0.567 \\ 1.267 \\ 0.067 \\ 1.973 \\ -0.567 \\ -0.607 \end{bmatrix}} \quad (9)$$

3. Chaining

With the connection between body $\mathbb{B}_1$ and $\mathbb{B}_2$ now changed from a rigid connection to a spherical hinge, a hinge map is introduced as of fig. 3.2 in [1]:

$$H = \begin{bmatrix} \mathbf{I}_3 \\ \mathbf{0}_3 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (10)$$

Then by using eq. 3.7 [1] the relative spatial velocity $\Delta_{\mathcal{V}}$ governing the two bodies is:

$$\Delta_{\mathcal{V}}(k) = H^T(k)\beta(k) \quad (11)$$

Where $\beta(k)$ denotes the generalized velocity coordinates vector across the body. Then by using eq. 3.19b [1] the velocity of a body k can be obtained:

$$\mathcal{V}(k) = \phi^T(k+1, k)\mathcal{V}(k+1) + \Delta_{\mathcal{V}}(k) \quad (12)$$

Which in this case becomes:

$$\mathcal{V}(\mathbb{B}_2) = \phi^T(\mathbb{B}_1, \mathbb{B}_2)\mathcal{V}(\mathbb{B}_1) + \Delta_{\mathcal{V}}(\mathbb{B}_2) \quad (13)$$

So for the case in Q 1.1.2 the relative angular velocity ${}^{\mathbb{B}_1}w(\mathbb{B}_1, \mathbb{B}_2)$ is given and a series of transformations from the spatial velocities of the point w in $\mathbb{B}_1$ are needed. They are the following and start with the transformation from point w to x in $\mathbb{B}_1$:

$${}^{\mathbb{B}_1}\mathcal{V}(\mathbb{I}, x) = \phi(w, x)^T {}^{\mathbb{B}_1}\mathcal{V}(\mathbb{I}, w) \quad (14)$$

From x the relative velocities of the two bodies are added as they are expressed in $\mathbb{B}_1$:

$$\mathbb{B}_1 \mathcal{V}(\mathbb{I}, x + \Delta_{\mathcal{V}}(\mathbb{B}_2)) = \mathbb{B}_1 \mathcal{V}(\mathbb{I}, w) + \mathbb{B}_1 \Delta_{\mathcal{V}}(\mathbb{B}_2) \quad (15)$$

This is followed by a transformation with only a difference in rotation between $\mathbb{B}_1$ and $\mathbb{B}_2$

$$\mathbb{B}_2 \mathcal{V}(\mathbb{I}, y) = \phi(x, y)^T \mathbb{B}_1 \mathcal{V}(\mathbb{I}, x + \Delta_{\mathcal{V}}(\mathbb{B}_2)) \quad (16)$$

Then a transformation from y to z which is only displaced by a length.

$$\mathbb{B}_2 \mathcal{V}(\mathbb{I}, z) = \phi(y, z)^T \mathbb{B}_2 \mathcal{V}(\mathbb{I}, y) \quad (17)$$

After the velocities are propagated to the outermost body, the velocities can then be expressed in the inertial fixed reference frame $\mathbb{I}$ using the same rotation as in eq. (9).

$$\mathbb{I} \mathcal{V}(\mathbb{I}, z) = \begin{bmatrix} \mathbb{I} \mathfrak{R}_{\mathbb{B}_2} & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbb{I} \mathfrak{R}_{\mathbb{B}_2} \end{bmatrix} \mathbb{B}_2 \mathcal{V}(\mathbb{I}, z) \approx \boxed{\begin{bmatrix} 3.5 \\ 3 \\ -0.2 \\ 2.853 \\ -2.167 \\ -1.327 \end{bmatrix}} \quad (18)$$

4. Equations of motion

The equations of motion (EoM) for the body is governed by eq. 2.28 [1] for a rigid body with forces and moments about an arbitrary point z in the body reference frame:

$$\mathbf{f}(z) = M(z) \dot{\beta}(z) + \mathbf{b}(z), \quad \mathbf{b}(z) = \bar{\mathcal{V}}(z) M(z) \mathcal{V}(z) \quad (19)$$

With $\mathbf{f}(z) = \begin{bmatrix} N(z) \\ F(z) \end{bmatrix} \in \mathbb{R}^6$ denoting the spatial forces. and the spatial inertia from eq. 2.7 [1]:

$$M(z) = \begin{bmatrix} \mathcal{J}(z) & m\tilde{p}(z) \\ -m\tilde{p}(z) & m\mathbf{I}_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6} \quad (20)$$

with eq. 2.11 [1]: $\mathcal{J}(z) = \mathcal{J}(c) - m\tilde{p}(z)\tilde{p}(z)$ and m being the mass of the body and $p = {}^{\mathbb{B}}\mathfrak{l}(z, C)$ describing the distance from z to C .

The $\overline{\mathcal{V}}(z)$ is the spatial vector cross-product in eq. 1.25 [1] of the $\mathcal{V}$ spatial velocities:

$$\overline{z} \triangleq \begin{bmatrix} \tilde{x} & \tilde{y} \\ \mathbf{0}_3 & \tilde{x} \end{bmatrix} \in \mathbb{R}^{6 \times 6} \quad (21)$$

As it is needed for the simulation to get the generalized accelerations $\dot{\beta}$ for each timestep, $\dot{\beta}$ is solved for:

$$\dot{\beta}(z) = M(z)^{-1}(\mathfrak{f}(z) - \mathfrak{b}(z)) = M(z)^{-1}(\mathfrak{f}(z) - \overline{\mathcal{V}}(z)M(z)\mathcal{V}(z)) \quad (22)$$

As $M(z)^{-1}$ is not dependent on time it can be precomputed and as such imposes a constant $\mathcal{O}(1)$ computational cost and not $\mathcal{O}(N)$ for N being the number of timesteps in the simulation.

It is also needed to get the time derivative of quaternions in eq. B.34 [1] when integrating over time in section 5:

$${}^{\mathbb{I}}\dot{\underline{q}}_{\mathbb{B}} = \frac{1}{2} \begin{bmatrix} -{}^{\mathbb{B}}\tilde{w} & {}^{\mathbb{B}}\tilde{w} \\ {}^{\mathbb{B}}\tilde{w}^T & 0 \end{bmatrix} \underline{q} \quad (23)$$

The linear velocity to the body from the Inertial fixed reference frame ${}^{\mathbb{I}}\mathfrak{l}_{\mathbb{B}}(\mathbb{I}, z)$ is:

$${}^{\mathbb{I}}\mathfrak{l}(\mathbb{I}, z) = {}^{\mathbb{I}}\mathfrak{R}_{\mathbb{B}} {}^{\mathbb{B}}\nu \quad (24)$$

The choice of generalized velocities is $\mathbb{I}(\mathbb{I}, z)$ which is the linear position from $\mathbb{I}$ to z as seen from $\mathbb{I}$.

The state vector can then be assembled:

$$s = \begin{bmatrix} \mathbb{I}\underline{q}_{\mathbb{B}} \\ \mathbb{I}(\mathbb{I}, z) \\ \beta_{\mathbb{B}} \end{bmatrix} \quad (25)$$

Then the derivative of the state with respect to time is obtained:

$$\dot{s} = \begin{bmatrix} \mathbb{I}\dot{\underline{q}}_{\mathbb{B}} \\ \mathbb{I}\dot{\mathbb{I}}(\mathbb{I}, z) \\ \dot{\beta}_{\mathbb{B}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \begin{bmatrix} -{}^{\mathbb{B}}\tilde{w} & {}^{\mathbb{B}}\tilde{w} \\ {}^{\mathbb{B}}\tilde{w}^T & 0 \end{bmatrix} \underline{q} \\ \mathbb{I}\mathfrak{R}_{\mathbb{B}}{}^{\mathbb{B}}\nu \\ M(z)^{-1}(\mathfrak{f}(z) - \overline{\mathcal{V}}(z)M(z)\mathcal{V}(z)) \end{bmatrix} \quad (26)$$

The equations of motions for the body is described within the state s with the derivative being $\dot{s}$.

5. Simulation

To simulate the system the states s and derivative $\dot{s}$ is integrated using numerical integration using a 4th order Runge-Kutta integration scheme (“RK45”) for 5 seconds with the relative and absolute tolerances of $1 \cdot 10^{-9}$ with a variable step solver using the Solve IVP method from SciPy and evaluated at 100 equidistant points. The initial conditions are given and with the bodyframe C coinciding and aligned with $\mathbb{I}$:

$$s_0 = [\mathbb{I}\underline{q}_{\mathbb{B}}^T \quad \mathbb{I}(\mathbb{I}, z)^T \quad \beta_{\mathbb{B}}^T]^T = [0 \ 0 \ 0 \ 1 \ 0.5 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]^T \quad (27)$$

The simulation results can be seen in Figures 2 and 3 below:

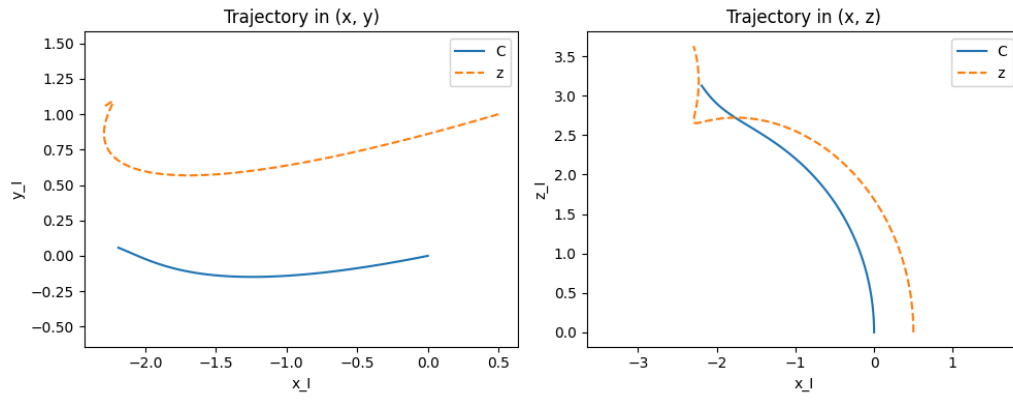


Figure 2: 2D plots of the trajectories of C and z viewed from the x-y (right) and x-z (left) planes with respect to the inertial frame

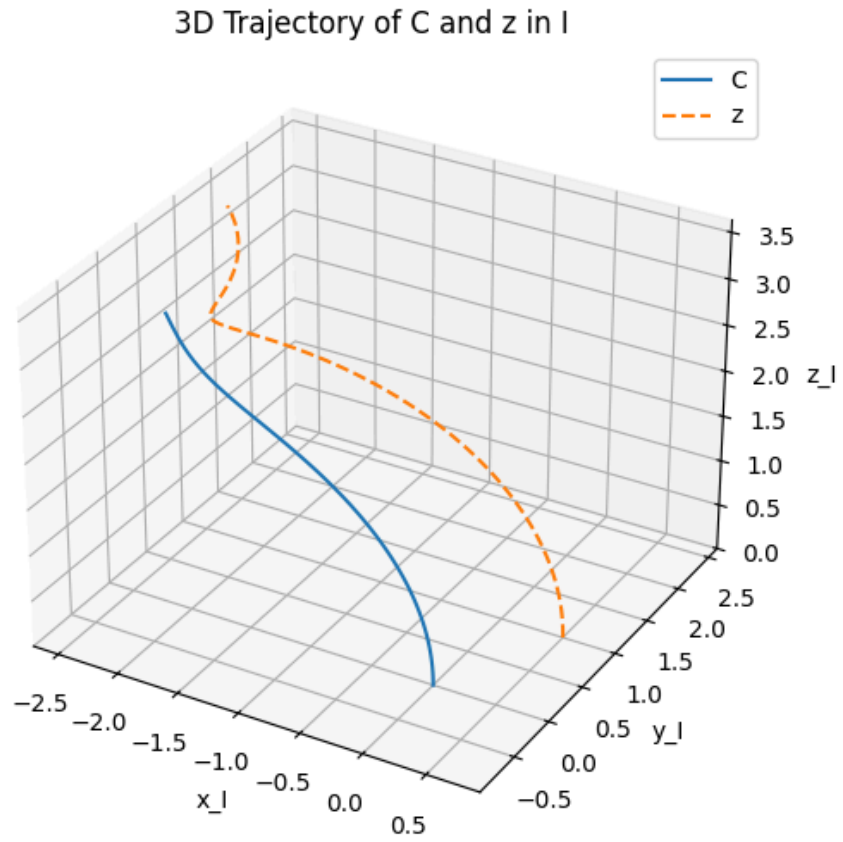


Figure 3: 3D plot of the trajectories of C and z as seen by I

References

- [1] A. Jain, Robot and multibody dynamics, 1st ed. 2011., Springer, New York, 2011.